

UNIT III Modelling in Business Intelligence

Models play a central role in Business Intelligence for achieving analysis goals

Models represent some part of the business process and allow precise formulation of interesting questions

Definition: A machine learning model is a mathematical representation of a real-world process created by training an algorithm on a dataset. It is used to make predictions or decisions without being explicitly programmed.

Examples:

- Classification Models: Predict categories or labels (e.g., spam detection in emails).
- Regression Models: Predict numerical values (e.g., house price prediction).

Need of models in BI

In Business Intelligence (BI), especially in tools like Power BI, data models are essential because they act as the foundation for accurate, efficient, and meaningful analysis.

Here's why models are needed in BI:

1. Organizing and Structuring Data

Raw data often comes from multiple sources (databases, Excel, APIs, etc.).

A data model organizes this data into tables, relationships, and hierarchies so it's easier to query and analyse.

2. Enabling Relationships between Data

Models define relationships (e.g., one-to-many, many-to-many) between tables.

This allows you to connect facts (sales, transactions) with dimensions (customers, products, dates) for deeper insights.

3. Improving Performance

Well-designed models **reduce redundancy** and optimize queries.

Aggregations and indexing in the model make **reports load faster**.

4. Supporting Calculations and KPIs

Models allow the use of DAX (Data Analysis Expressions) to create calculated columns, measures, and KPIs.

This ensures consistent business logic across all reports.

5. Ensuring Data Accuracy and Consistency

Centralized models enforce single sources of truth.

Everyone in the organization works with the same definitions for metrics like revenue, profit, or churn rate.

6. Enabling Scalability and Reusability

Once built, a model can be reused across multiple dashboards and reports.

This saves time and ensures consistency in analytics.

In short:

A BI model transforms raw, scattered data into a structured, connected, and optimized dataset that supports accurate, fast, and consistent decision-making.

Role in decision-making

Modelling in Business Intelligence (BI) involves the use of various mathematical, statistical, and computational techniques to transform data into actionable insights. These models **help in understanding past performance, predicting future outcomes, and making informed decisions**

Abstraction in Modelling

Abstraction means removing unnecessary details while retaining important features.

Techniques of Abstraction

1. Simplification

Removing minor variables that do not significantly affect outcomes.

2. Aggregation

Combining similar elements into groups.

Example: Customers grouped as “Retail” and “Wholesale.”

3. Idealization

Assuming ideal conditions such as perfect competition or constant demand.

Example:

Suppose a farmer sells wheat in a market. Under idealization, we assume:

Hundreds of farmers sell identical wheat

Buyers can freely choose from any seller

The market decides the price

In reality, some farmers may have better quality wheat or transport advantages, but the model ignores these factors to simplify analysis.

4. Generalization

Creating general rules from specific cases.

5. Assumptions

Making logical assumptions to limit complexity.

Example

In a production planning model:

$\text{Profit} = (\text{Selling Price} \times \text{Quantity}) - \text{Total Cost}$

Factors like worker mood are ignored for simplification.

Advantages

1. Easier problem-solving

2. Cost-effective analysis

3. Improves forecasting

Limitations

1. Oversimplification may reduce accuracy

2. Wrong assumptions lead to incorrect results

Conclusion

Representation and abstraction are foundational in modelling as they convert complex real-life problems into manageable analytical forms.

Model Presentation

Creating an effective model presentation in Business Intelligence (BI) involves communicating complex data and insights in a clear, concise, and actionable manner. Here's a structured approach to presenting a BI model:

Steps in Model creation

Introduction

Objective: Clearly state the purpose of the model. What problem does it aim to solve or what question does it answer?

Scope: Define the scope of the analysis. Mention the data sources and the time frame covered.

Data Overview

Data Sources: Describe where the data was obtained (e.g., internal databases, external APIs, spreadsheets).

Data Preparation: Outline the steps taken to clean and pre-process the data, including any transformations or aggregations.

Model Selection

Model Type: Explain the type of model used (e.g., regression, classification, clustering).

Justification: Provide reasons for choosing this particular model. Highlight its suitability for the problem.

Methodology

Model Building: Detail the steps taken to build the model, such as feature selection, training, and validation processes.

Algorithms and Techniques: Mention any specific algorithms, statistical methods, or machine learning techniques used.

Results

Key Metrics: Present the key performance metrics of the model (e.g., accuracy, precision, recall, R-squared).

R-squared is a statistical measure that represents the goodness of fit of a regression model. The value of R-square lies between 0 to 1. Where we get R-square equals 1 when the model perfectly fits the data and there is no difference between the predicted value and actual value

Visualizations: Use charts, graphs, and tables to visually represent the results. Highlight the most important findings.

Insights

Interpretation: **Provide an interpretation of the results.** What do the findings mean in the context of the business problem?

Implications: **Discuss the implications of the results for the business.** How can these insights be used to drive decision-making?

Recommendations

Actionable Steps: Offer concrete recommendations based on the model's insights. **What actions should be taken?**

Future Work: Suggest areas for further analysis or model improvement. Mention any additional data sources that could be leveraged.

Conclusion

Summary: Recap the main points of the presentation, reiterating the objective and how the model addresses it.

Questions: Open the floor for questions and discussion. Encourage feedback and engagement from the audience.

Example Visualization

Here's an example layout for a presentation slide on a **predictive sales model**:

Section	Content
Introduction	Objective: Predict future sales based on historical data.
Data Overview	Data Sources: Sales databases, market reports. Data Preparation: Cleaning, normalization.
Model Selection	Model Type: Linear Regression. Justification: Simplicity and interpretability.
Methodology	Feature selection, training on 80% of data, validation on 20% of data.
Results	Key Metrics: R-squared = 0.85, RMSE = 10,000. R squared is a statistical measure of how well a regression model predicts the dependent variable.

The root mean square error (RMSE) measures the average difference between a statistical model's predicted values and the actual values

Insights Interpretation: Strong correlation between marketing spend and sales.

Recommendations Increase marketing budget in underperforming regions.

Conclusion Summary of findings and next steps.

Techniques for Presenting Models (Visual, Mathematical, Textual)

Presenting models effectively involves a combination of visual, mathematical, and textual techniques. Here are some key strategies to consider:

Visual Models: Use charts, graphs, and diagrams to represent data visually. Ensure the visuals are clear, concise, and relevant to the model being presented.

Mathematical Models: Present mathematical concepts and calculations in a way that is understandable to the audience. Use clear explanations and examples to illustrate the model's principles.

Textual Models: Use clear and concise text to describe the model's structure and purpose. Use language that is accessible to the audience.

By employing these techniques, you can create a presentation that effectively communicates the model's content and engages the audience. Remember to balance the use of each type of model to create a cohesive and informative presentation

Model Assessment

Assessing the quality of models in Business Intelligence (BI) is crucial to ensure they are reliable, accurate, and actionable. Here are the key aspects of model assessment and quality evaluation in BI:

Model Assessment

- **Performance Metrics:**

Accuracy: Measures how often the model makes correct predictions.

Precision: The ratio of **true positive predictions** to the **total positive predictions**.

Recall: The ratio of **true positive predictions to the actual positives**.

Below is the confusion matrix for binary classifier predicting a present of disease

There are **total 150 predictions**.

Out of 150 cases, the classifier **predicted “Yes” 100 times** and **“No” 50 times**

	Predicted No	Predicted Yes
Actual No	45	5
Actual Yes	5	95

- **Accuracy**
- **Precision**
- **Recall**
- **F1-Score**

- **Accuracy:** Overall, how often is the classifier correct?

$$\begin{aligned}
 \text{Accuracy} &= \frac{TN+TP}{TN+FP+FN+TP} \\
 &= \frac{45 + 95}{150} = \underline{93.33\%}
 \end{aligned}$$

	Predicted No	Predicted Yes
Actual No	TN = <u>45</u>	FP = <u>5</u>
Actual Yes	FN = <u>5</u>	<u>TP = 95</u>

- **True Positive Rate:** When it's actually yes, how often does it predict yes?
- also known as "Sensitivity" or "Recall"

$$\begin{aligned}
 \text{True Positive rate} &= \frac{TP}{\text{Actual Yes}} \\
 &= \frac{95}{100} = \underline{95\%}
 \end{aligned}$$

	Predicted No	Predicted Yes
Actual No	TN = 45	FP = 5
Actual Yes	FN = 5	<u>TP = 95</u>

- **Precision:** When it predicts yes, how often is it correct?

$$\text{Precision} = \frac{TP}{\text{Predicted Yes}}$$

$$= \frac{95}{100} = 95\%$$

	Predicted No	Predicted Yes
Actual No	TN = 45	FP = 5
Actual Yes	FN = 5	TP = 95

F1 Score: The F1 score is the harmonic mean of precision and recall, providing a balance between the two,

Unlike the arithmetic mean, which simply sums up the values and divides by the count, the harmonic mean calculates the reciprocal of the average of the reciprocals of the values.

Example Calculation

Let's calculate the harmonic mean of the numbers 3, 4, and 6:

1. Reciprocals:

$$\begin{aligned} 1/3 &= 0.333, \\ 1/4 &= 0.25, \\ 1/6 &= 0.167 \end{aligned}$$

2. Sum of Reciprocals:

$$0.333 + 0.25 + 0.167 = 0.75$$

3. Number of Data Points:

$$n = 3$$

4. Harmonic Mean Calculation:

$$HM = 3 / 0.75 = 4$$

R-Squared: Indicates how well the model explains the variance in the dependent variable.

Root Mean Square Error (RMSE): Measures the average magnitude of prediction errors.

- **Validation Techniques:**

1. Cross-Validation: Splitting the data into multiple folds and validating the model on each fold to ensure it generalizes well.

Example of k-Fold Cross-Validation

Let's say we have a dataset with 1000 samples and we want to use 5-fold cross-validation:

The dataset is split into 5 folds, each containing 200 samples.

The model is **trained on 4 folds** (800 samples) and **validated on the remaining fold (200 samples)**.

This process is repeated 5 times, with each fold used as the validation set once.

The performance metrics (e.g., accuracy, precision, recall) from each fold are averaged to get the final performance estimate.

2. Train-Test Split: Dividing the data into a training set and a test set to evaluate model performance on unseen data.

Cross-validation is a powerful technique for evaluating machine learning models and ensuring they perform well on new data. By using multiple subsets of the data for training and validation, it provides a more accurate and robust estimate of model performance

- **Overfitting and Underfitting:**

Overfitting: When a model performs **well on training data** but **poorly on new**, unseen data. It indicates the model is too complex.

Underfitting: When a model performs **poorly on both training and test data**. It indicates the model is too simple.

- **Confusion Matrix:**

A table used to evaluate the performance of classification models. It shows the true positives, true negatives, false positives, and false negatives. **A confusion matrix is a table that shows where your classification model gets predictions right and where it fails.**

- **ROC Curve and AUC:**

ROC Curve: A graphical **representation of a model's true positive rate vs. false positive rate.**

A **Receiver Operating Characteristic (ROC) curve** is a graphical plot that illustrates the performance of a binary

classifier model at various threshold settings. **It is used to visualize the trade-off between the true positive rate (TPR) and the false positive rate (FPR) across different thresholds**

AUC (Area Under the Curve): **Measures the overall performance of the model.** A higher AUC indicates better model performance.

Quality of Models

Quality Criteria for Business Process Models

1. Correctness: **The model should be syntactically correct** (This means that the model must be constructed using the correct elements and relationships as defined by the modelling standard). Within the used model structure, and it combines the model semantics (The meaning of the elements and their relationships within the model) with the domain semantics (the meaning of the concepts and processes within the specific business domain) in an appropriate way.

Let's look at a **simple example of model semantics** in healthcare using a **predictive model for identifying high-risk patients for diabetes**. This example will illustrate the basic principles of model semantics, including feature interpretation, output interpretation, and contextual relevance.

Example: Predictive Model for Diabetes Risk

Scenario: A healthcare provider wants to identify patients who are at high risk of developing diabetes within the next year. They use a predictive model based on various patient features.

1. Feature Interpretation

- **Age:**
 - **Meaning:** The patient's age in years.
 - **Impact:** Older patients may have a higher risk of developing diabetes due to age-related health factors.
- **Body Mass Index (BMI):**
 - **Meaning:** A measure of body fat based on height and weight.
 - **Impact:** Higher BMI values are associated with an increased risk of diabetes.
- **Family History of Diabetes:**
 - **Meaning:** Whether the patient has close relatives (e.g., parents, siblings) with diabetes.
 - **Impact:** A positive family history increases the likelihood of developing diabetes.
- **Physical Activity Level:**
 - **Meaning:** How often the patient engages in physical exercise (e.g., sedentary, moderate, active).
 - **Impact:** Lower physical activity levels are associated with a higher risk of diabetes.
- **Blood Glucose Level:**
 - **Meaning:** The concentration of glucose in the patient's blood.
 - **Impact:** Elevated blood glucose levels can indicate prediabetes or diabetes.

2. Model Output Interpretation

- **Diabetes Risk Score:**
 - **Meaning:** A numerical score representing the likelihood that a patient will develop diabetes within the next year.
 - **Example:** A diabetes risk score of 0.85 means the model predicts an 85% chance that the patient will develop diabetes within the next year.

3. Contextual Relevance

- **Example Scenario:**

- A 60-year-old patient with a BMI of 30, a family history of diabetes, low physical activity, and elevated blood glucose levels has a diabetes risk score of 0.85.
- **Action:** The healthcare provider may recommend lifestyle changes, regular monitoring, and preventive measures to reduce the patient's risk of developing diabetes.

Conclusion

By understanding the semantics of the predictive model, healthcare providers can interpret the model's features and outputs meaningfully and take appropriate actions to manage and reduce the risk of diabetes in high-risk patients. This ensures that the insights provided by the model are aligned with the real-world context and support effective decision-making in healthcare.

Model semantics are essential for ensuring that models are meaningful, accurate, and useful in their intended application

Example: Domain Semantics in Healthcare

Consider the healthcare domain:

- **Concepts and Entities:**
 - Patient
 - Doctor
 - Diagnosis
 - Treatment
- **Relationships:**
 - A "Doctor" diagnoses a "Patient."
 - A "Patient" receives a "Treatment."
- **Attributes and Properties:**
 - A "Patient" has attributes like "Name," "Age," and "Medical History."
 - A "Treatment" has properties like "Medication" and "Dosage."
- **Rules and Constraints:**
 - A "Treatment" must be prescribed by a "Doctor."

- A "Diagnosis" must be based on "Medical Tests."
- **Taxonomy and Hierarchy:**
 - Medical Professional (parent class)
 - Doctor (subclass)
 - Nurse (subclass)

By defining the domain semantics, the healthcare system can accurately represent and interpret the data, ensuring consistency and clarity across various applications and systems.

2. Relevance: The model has to comply with its intended function. In BI, relevance usually **depends on the ability to explain past observations and to predict future observations.**

3. Economic efficiency: There should be a **reasonable trade-off between the model complexity** necessary for achieving the implied needs and the **cost of usage of the model** in practice.

It involves making decisions that maximize the benefits while minimizing the costs, ensuring that resources are allocated in the most effective way.

4. Clarity: The model should be **understood by the users of the model** and should have some aesthetic appeal (**create dashboards, reports, and visualizations that are not only functional and informative but also pleasing to the eye.**).

5. Comparability: The model should **fit into the overall framework** of an analysis of a business process. In particular, if **we use different models for instances of the same business process, the results should be consistent.**

Scenario: The company notices a significant increase in order delays during peak seasons.

Descriptive Model: Analyzes past peak seasons and identifies that processing times tend to increase by 20% due to higher order volumes.

Predictive Model: Forecasts that the upcoming peak season will see a 25% increase in orders, potentially leading to even longer processing times if no changes are made.

Prescriptive Model: Recommends increasing staff levels during peak seasons and optimizing warehouse layout to streamline order processing.

Consistency Check:

- The descriptive model's insights on past performance should align with the predictive model's forecasts.
- The prescriptive model's recommendations should be based on the trends and forecasts provided by the other models.

6. Data Quality:

The accuracy and completeness of the data used for training the model significantly impact its quality. **Ensure data is clean, relevant, and representative of the problem domain.**

7. Model Interpretability:

The ability to understand and interpret the model's predictions. Transparent models (like linear regression) are often preferred in BI for their interpretability.

8. Robustness:

The model's **ability to handle various types of data and perform well under different conditions.** Robust models are less sensitive to outliers and noise in the data.

9. Scalability:

The model's **ability to handle large datasets and scale with increasing data volume without significant performance degradation.**

10.Timeliness:

The time it takes for the model to process data and generate predictions. In BI, timely insights are crucial for decision-making.

11.Business Value:

The practical impact of the model's insights on business decisions. **Models should provide actionable insights** that align with business goals and objectives.

Best Practices for Model Quality Assurance

Regular Monitoring: **Continuously monitor model performance** and update it as needed to ensure it remains accurate and relevant.

Documentation: **Maintain detailed documentation** of the model's development process, including data sources, assumptions, and validation methods.

Collaboration: **Involve domain experts and stakeholders** in the model development process to ensure the model addresses the right questions and provides valuable insights.

Testing: Rigorously **test models under different scenarios** to identify potential weaknesses and improve their robustness.

By rigorously assessing and maintaining the quality of BI models, organizations can ensure they derive reliable insights and make data-driven decisions.

Logical Structures

Logical structures are frameworks used in mathematical logic and model theory to represent formal systems. They consist of a domain of objects, relations, and operations that interpret the symbols of a formal language. Logical structures allow the study of consistency, completeness, and decidability of theories, and they are central to understanding how axioms define a system. For example, the natural numbers with

standard arithmetic operations form a logical structure for arithmetic statements ().

Algebraic Models

Algebraic models are mathematical structures that generalize algebraic systems such as groups, rings, fields, and vector spaces. In model theory, an algebraic model is a structure that satisfies a set of axioms defining a particular algebraic system. These models help classify algebraic systems, study their properties, and explore relationships between abstract logic and concrete algebraic systems. Applications include cryptography, computer science, and theoretical mathematics ().

Graph models represent systems as nodes (vertices) and edges (connections), capturing relationships and interactions between entities. They are widely used in network analysis, computer science, and social sciences. Graph models can be directed or undirected, weighted or unweighted, and are useful for modeling communication networks, transportation systems, and dependency structures. They provide a visual and computational framework for analyzing connectivity, flow, and structural properties.

Analytical Models

Analytical models use mathematical equations and functions to describe the behavior of systems quantitatively. These models are common in data science, physics, and engineering, where they predict outcomes based on input variables. Analytical models often involve statistical, probabilistic, or deterministic methods to simulate real-world phenomena, optimize

processes, or forecast trends. They are essential for decision-making and scenario analysis ().

Logical structures, such as ontology and frame, play a crucial role in Business Intelligence (BI) modelling by providing a framework for organizing and representing data. Here's how they are used:

Ontology:

Definition: Ontology **defines the relationships and properties of entities within a domain**. In BI modelling, **ontologies are used to create a formal representation of the data**, including the entities, their attributes, and the relationships between them.

By using ontologies, BI models can **ensure consistency and standardization in data representation, making it easier to understand and analyse complex business data**.

Key Components of Ontology

Classes (Concepts): Represent abstract groups or categories of objects within the domain. For example, Person or Vehicle.

Instances (Individuals): Concrete examples or occurrences of classes, such as JohnDoe as an instance of Person.

Attributes (Properties): Describe characteristics or features of classes or instances; for example, a Person may have an attribute age.

Relationships: Define how classes or instances are related to each other, such as hasChild between two Person instances.

Constraints and Rules: Restrictions placed on classes, properties, or relationships (e.g., a person can have only one birthdate).

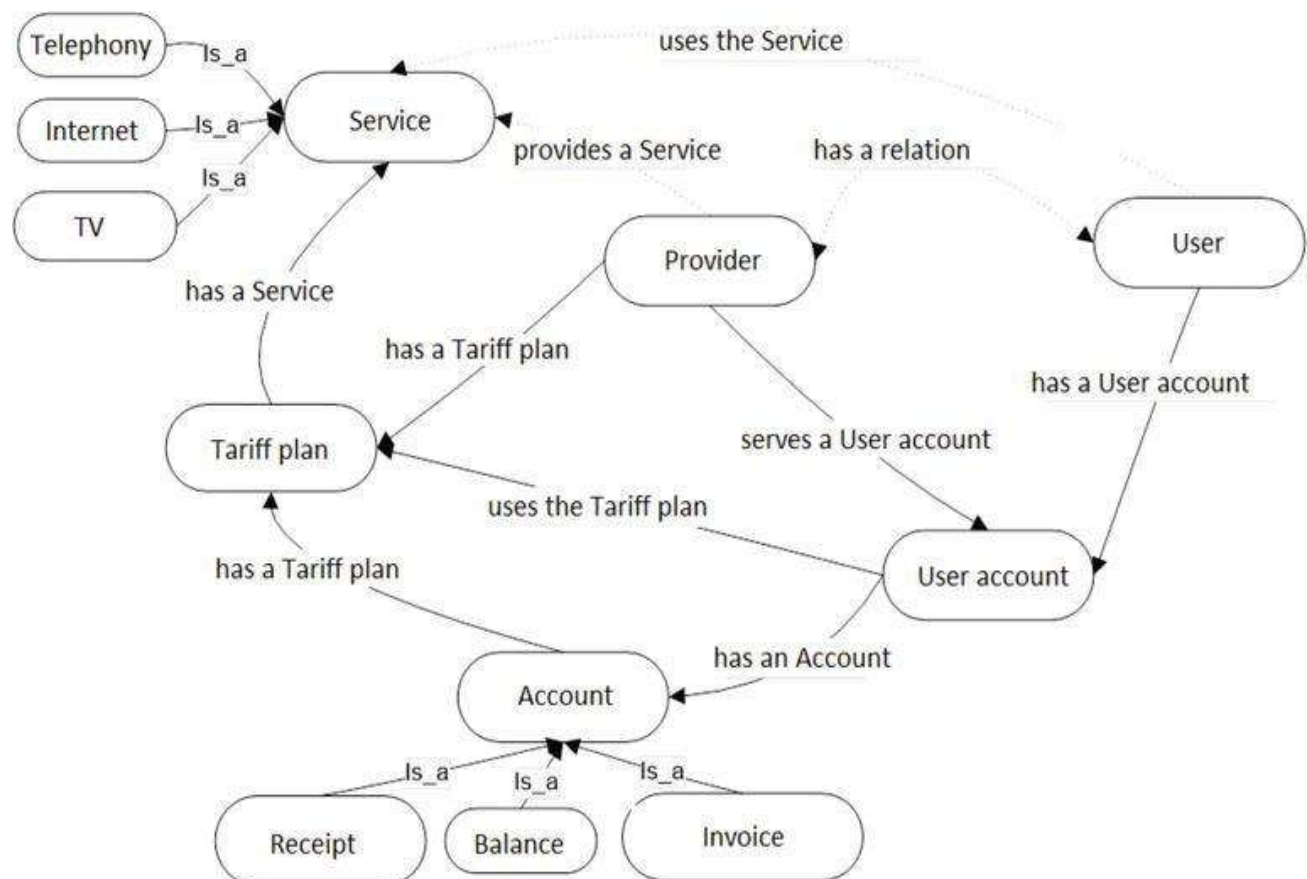
Relationships in Ontology

Ontologies represent various kinds of relationships including:

Is-a (Subclass): Represents inheritance or class hierarchy (e.g., a Car is-a Vehicle).

Part-of: Indicates composition or component relationships (e.g., an Engine is part-of a Car).

Association: General links between classes or instances (e.g., Person works-for Company).



The class “User” is associated with the “Account” class with the relation “has an Account”

The class “Provider” is associated with the “Service” class with the “provides a Service” relation

The “Tariff Plan” class is related to the “Service” class with the “has Service” relation

The “Account” class is associated with the “Tariff Plan” class

The “Paid Invoice (Receipt)” class has the following data type relations:

- “has a paid amount” with the data type integer (paid amount)
- “has a payment date” with the dateTime (date of payment for this account)

The “Balance” class has the following data type relations:

- “has a sum” with the type of data integer (amount of money on the balance sheet)
- “has a balance date” with the dateTime (date for which the balance information is viewed).

Ontology: Focuses on defining a shared vocabulary and the **relationships between concepts within a domain.**

Ontologies provide a **high-level, shared understanding of a domain**, while **frames offer detailed, structured representations** of specific objects or situations.

Frame: Focuses on **representing knowledge about specific objects** or situations, organizing their attributes and values.

Frame:

In BI modelling, frames can be used to define the structure and behaviour of business entities.

Example Frame: Customer Interaction

Frame Name: Customer Interaction (Customer, Interaction, and Frame are the entities)

Slots:

Customer Information:

CustomerID: Unique identifier for the customer.

Name: Full name of the customer.

Age: Age of the customer.

Address: Residential address of the customer.

Interaction Details:

InteractionType: Type of interaction (e.g., Purchase, Inquiry, Complaint).

Date: Date of the interaction (Format: YYYY-MM-DD).

Channel: The medium through which the interaction occurred (e.g., In-Store, Online, Phone).

Product Information (if applicable):

ProductID: Unique identifier for the product.

ProductName: Name of the product.

Category: Category of the product.

Price: Price of the product.

Outcome:

ResolutionStatus: Status of the interaction outcome (e.g., Resolved, Pending, Escalated).

Feedback: Customer feedback or comments on the interaction.

FollowUpRequired: Boolean flag indicating if a follow-up is needed (e.g., Yes, No).

Example Instance of Customer Interaction Frame

Frame: Customer Interaction

Slots:

- Customer Information:

- CustomerID: C12345

- Name: John Doe

- Age: 35

- Address: 123 Main St, Anytown, USA

- Interaction Details:

- InteractionType: Purchase
- Date: 2024-12-01
- Channel: Online

- Product Information:
 - ProductID: P67890
 - ProductName: Smartwatch
 - Category: Electronics
 - Price: \$199.99

- Outcome:
 - Resolution Status: Resolved
 - Feedback: Very satisfied with the purchase and delivery time.
 - FollowUp Required: No

Explanation:

Customer Information: Captures key details about the customer, such as their ID, name, age, and address.

Interaction Details: Includes specific information about the interaction, such as its type, date, and channel.

Product Information: If the interaction involves a product, this section captures relevant product details.

Outcome: Summarizes the result of the interaction, including the resolution status, customer feedback, and whether a follow-up is required.

Benefits:

Clarity: Frames provide a clear, structured way **to represent complex information**, making it easier to understand and analyze.

Context: By encapsulating related attributes and their values, frames add context to the data, leading to more accurate interpretations.

Decision Support: Frames **enhance the decision-making process** by providing relevant, contextual information.

This example demonstrates how frames can be used to model and organize customer interaction data in BI, leading to better insights and more informed business decisions.

Applications of Frame

Expert systems, Artificial intelligence, cognitive science, and knowledge representation.

Modelling using graph structure: Business process model and notation (BPMN)

[BPMN Diagram Symbols & Notation | Lucidchart](#)

Like flowcharts, BPMN diagrams use a set of standard symbols. Each shape has a specific meaning and business context where it's most appropriate.

A business process model and notation diagram, or BPMN diagram for short, is **used to build easy-to-read business process model** flowcharts, which can be shared across organizations and industries.

BPMN diagram symbols are categorized into four main groups:

Flow objects,

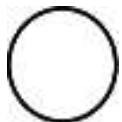
Connecting objects,

Swimlanes,

Artifacts

BPMN event types

Events represent an event in a business process.



Start event symbol - Signals the first step of a process



Intermediate event symbol - Represents any event that occurs between a start and end event



End event symbol - Signals the final step in a process

BPMN event symbols

Each of these BPMN event symbols can be styled to represent the specific details of their process. The examples below are contained within start event symbols, but they can be combined with any event type. Common event symbols represent the following circumstances:



Message symbol - Triggers the process, facilitates intermediate processes, or finishes the process.



Timer symbol - A time, date, or recurring time and date triggers the process, aids intermediates processes, or completes the process.



Escalation symbol - A step reacts on an escalation and flows to another role in the organization. This event is only used within an event sub-process. **An escalation occurs when someone at a higher level of responsibility within the organization becomes involved in a process.**



Conditional symbol - A process begins or continues when a business condition or business rule is met



Link symbol - A sub-process that is part of a larger process

 Error symbol - A caught error at the start, middle, or end of a process. An event sub-process with an error trigger will always interrupt its containing process

 Cancel symbol - Reacts on a transaction that was canceled within a sub-process. In an end event, the cancel symbol represents the triggered cancellation of a process

 Compensation symbol - A refund that's triggered when operations partially fail.

 Signal symbol - A signal that communicates across different processes. A signal symbol can begin a process, facilitate it, or complete it.

 Multiple symbol - Multiple triggers initiating a process

 Parallel multiple symbol - A process instance that doesn't start, continue, or end until all possible events have occurred.

 Terminate symbol - Triggers the immediate termination of a process step. All related instances are terminated at the same time

BPMN activity symbols



Task symbol - The most basic level of an activity and cannot be broken down further. For example, a morning routine process might involve the task of turning on your computer.



Transaction symbol - Is a specialized sub-process that involves payment.



Call symbol - Is a global sub-process that is reused at various points in the business flow.



Sub-process symbol - Is a group of tasks that fit together particularly well

BPMN gateway symbols

In a BPMN diagram, gateway symbols that separate and recombine flows. There are several types of gateways:



Exclusive symbol - Evaluates the state of the business process and, based on the condition, breaks the flow into one or more mutually exclusive paths. For example, one report will be written if a supervisor grants approval; no report will be generated if the supervisor withholds approval.



Event-based symbol - An event-based gateway is similar to an exclusive gateway—both involve one path in the flow. In the case of an event-based gateway, however, you are evaluating which event has occurred, not which condition is being met. **For example, you might wait to send an email until the CEO has arrived at the office. If the CEO doesn't arrive, the email will remain unsent**



parallel gateways are used to represent two concurrent tasks in a business flow. An example is a marketing department generating new sales leads and contacting existing leads at the same time.



Inclusive symbol - **Breaks the process flow into one or more flows.** For example, an inclusive gateway could involve business actions taken based on survey results. One process may be triggered if the consumer is satisfied with Product A. Another flow is triggered when the consumer indicates that they are satisfied with product B, and a third process is triggered if they aren't satisfied with A.



Parallel event-based symbol - As the name suggests, this gateway is similar to a parallel gateway. **It allows for multiple processes to happen at the same time, but unlike the parallel gateway, processes are event-dependent.**

Connecting objects in a BPMN diagram

Connecting objects are lines that connect BPMN flow objects. There are three different types: sequence flows, message flows, and associations.

-  Sequence flow symbol - **Connects flow objects in proper sequential order.**
-  Message flow symbol - **Represents messages from one process participant to another.**
-  Association symbol - **Shows relationships between artifacts and flow objects.**

Swimlanes in a BPMN diagram

Swimlanes are **used to organize aspects of a process** in a BPMN diagram. **Swimlanes visually group objects into lanes, with each aspect of the process added to a separate lane. These elements can be arranged either horizontally or vertically.** Swimlanes not only organize activities into separate categories, they can reveal delays, inefficiencies, and the workers responsible for each step in a process.



Artifacts in a BPMN diagram

Artifacts represent information relevant to the model but not to individual elements within the process.

The three artifact types are

annotations, groups, and data objects that can be used in a BPMN diagram. All three are used to augment and describe a BPMN process.

Annotations allow the modeler to describe additional flow parts of the model or notation.

Groups organize tasks or processes that have significance in the overall process.

Data objects represent data placed into the process, data resulting from the process, data that needs to be collected, or data that must be stored.





Data input symbol - **Represents data requirements** that tasks in the business process depend on.



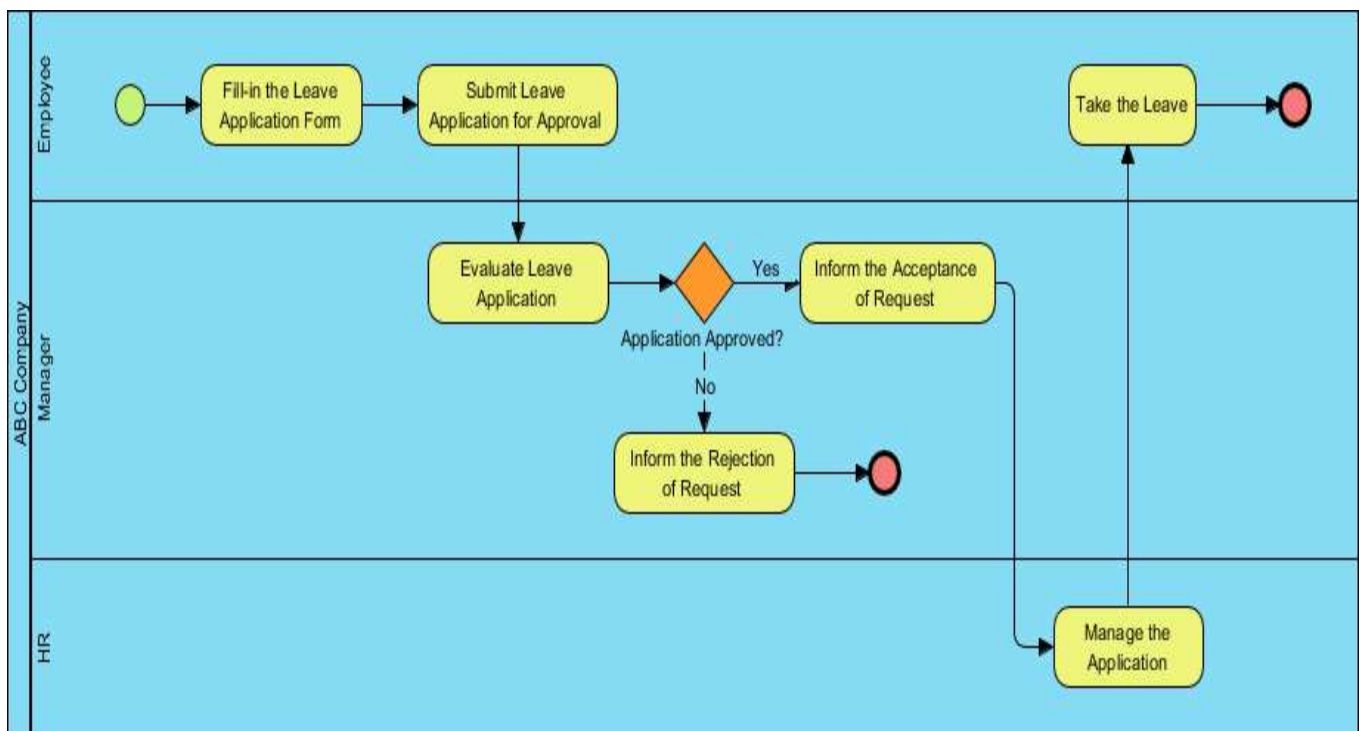
Data output symbol - **Demonstrates information produced** as the result of a business process.



Data collection symbol - **Signifies information collected** within a business process.



Data storage symbol - Represents the **ability to store or access data** that's associated with a business process



For more details you may refer [BPMN Activity Types Explained](#)

Modelling using probabilistic structures

Modelling using probabilistic structures

Modeling using probabilistic structures involves using probability theory to represent uncertainty and make predictions based on incomplete or uncertain information. Here are some key probabilistic structures and how they can be applied in various contexts:

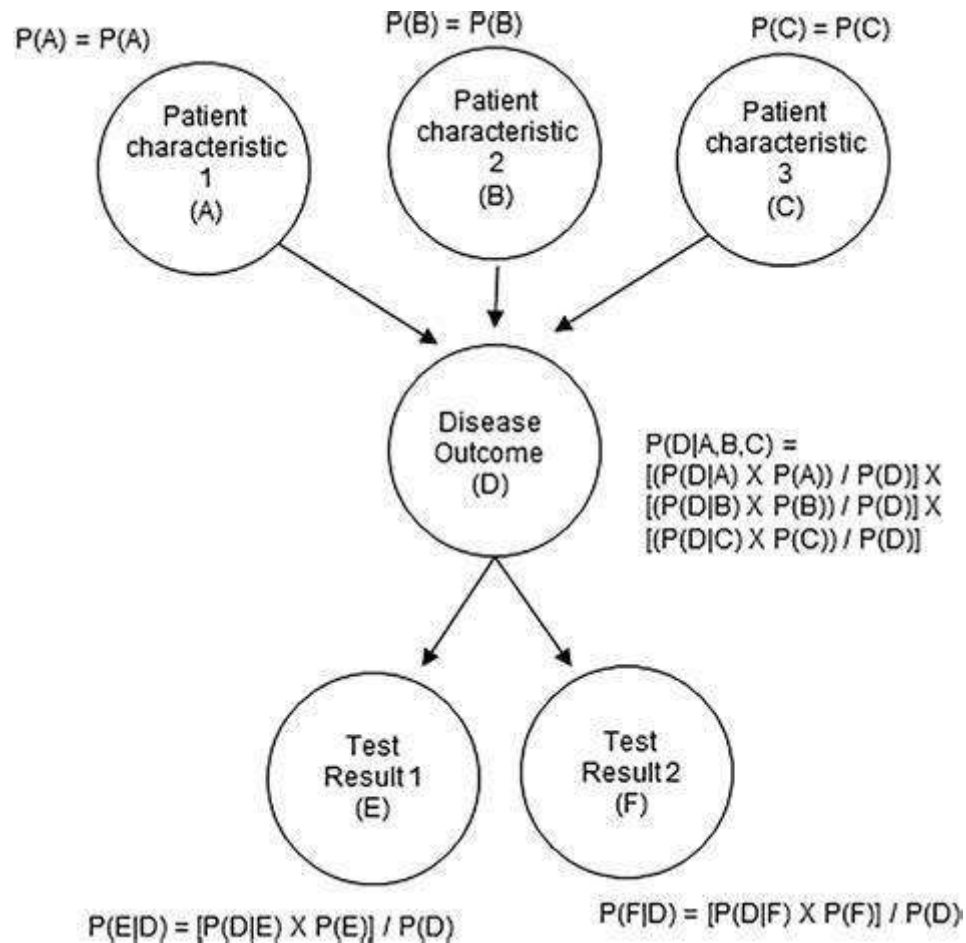
1. Bayesian Networks

Overview:

Bayesian Networks are graphical models that represent the probabilistic relationships among a set of variables using directed acyclic graphs (DAGs). Each node represents a variable, and the edges represent conditional dependencies.

Example: Medical Diagnosis

- **Variables: Disease (D), Symptoms (S), Test Results (T)**
- **Objective: To diagnose a disease based on observed symptoms and test results.**
- **Approach:**
 - Nodes represent the presence of a disease, symptoms, and test results.
 - Edges represent the causal relationships between these variables.
 - Conditional Probability Tables (CPTs) define the probability of each variable given its parents.



In this example, the arrows represent dependencies between the variables.

We can see that three patient characteristics influence the likelihood of disease, and that the status of these patient characteristics can propagate through the network to influence test results for either test 1 or test 2.

Using this example, we see that, unlike other machine-learning approaches such as artificial neural networks, **users can clearly understand how a particular probability was determined and which variables are contributing proximally and distally to the outcomes of interest, that is, their relative position in the network.**

How It Works:

- **Inference: Given observed symptoms and test results, the Bayesian Network updates the probability of having a specific disease.**
- **Example: If a patient has a fever and a positive test result, the Bayesian Network calculates the updated probability of the patient having the disease.**

2. Markov Chains

Overview:

Markov Chains are mathematical models that describe systems transitioning between states with certain probabilities. The key feature is that the future state depends only on the current state (memory less property).

Example: Weather Prediction

- **States: Sunny, Rainy, Cloudy**
- **Objective: To predict the weather for the next day.**
- **Approach:**
 - **Define the states (e.g., Sunny, Rainy, Cloudy).**
 - **Establish transition probabilities (e.g., the probability of transitioning from Sunny to Rainy).**

How It Works:

- **Transition Matrix: Represents the probabilities of moving from one state to another.**
- **Prediction: Use the current state and the transition matrix to predict the next state.**
- **Example: If today is Sunny, the Markov Chain can predict the probability of it being Sunny, Rainy, or Cloudy tomorrow.**

3. Hidden Markov Models (HMMs)

Overview:

HMMs are used to model systems where the states are not directly observable (hidden). They are particularly useful for modelling sequences of events.

Example: Speech Recognition

- **States: Phonemes (basic units of sound)**
- **Observations: Acoustic signals (features extracted from audio)**
- **Objective: To convert spoken words into text.**
- **Approach:**
 - **Define hidden states (phonemes) and observations (acoustic signals).**
 - **Use transition probabilities and emission probabilities (probability of observing a signal given a state).**

How It Works:

- **Viterbi Algorithm: Finds the most probable sequence of hidden states given the observations.**
- **Example: Given an audio clip, the HMM decodes the sequence of phonemes and converts it into text.**

4. Monte Carlo Simulation

Overview:

Monte Carlo Simulation is a computational technique that uses repeated random sampling to estimate the probability of different outcomes.

Example: Financial Risk Assessment

- **Variables: Stock prices, Interest rates, Market conditions**
- **Objective: To assess the risk of an investment portfolio.**
- **Approach:**
 - **Define the variables and their probability distributions.**
 - **Run simulations to generate possible outcomes.**

How It Works:

- **Random Sampling: Generate random values for the variables based on their distributions.**
- **Simulation: Calculate the outcomes for each set of random values.**
- **Example: Run simulations to estimate the probability of the portfolio value falling below a certain threshold.**

Conclusion

Probabilistic structures provide powerful tools for modeling uncertainty and making data-driven decisions in various fields. By leveraging Bayesian Networks, Markov Chains, Hidden Markov Models, and Monte Carlo Simulation, we can gain valuable insights, predict future outcomes, and optimize processes

Logical structures, such as ontology and frame

1. What is a model in Business Intelligence?

- A) A raw data collection system**
- B) A simplified representation of real-world processes**
- C) A database management tool**
- D) A type of programming language**

Answer: B) A simplified representation of real-world processes

2. The primary purpose of using models in BI is to:

- A) Store large volumes of data**
- B) Replace managers**
- C) Support analysis and decision-making**
- D) Eliminate uncertainty completely**

Answer: C) Support analysis and decision-making

3. Models help decision-makers by:

- A) Increasing data complexity**
- B) Providing structured insights from data**
- C) Avoiding data usage**
- D) Replacing human judgment entirely**

Answer: B) Providing structured insights from data

4. Which of the following best describes a BI model?

- A) A random data summary**
- B) A theoretical framework used to analyze business situations**
- C) A spreadsheet with no formulas**
- D) A hardware component**

Answer: B) A theoretical framework used to analyze business situations

5. Why are models needed in Business Intelligence?

- A) To make systems more complicated**
- B) To simplify complex business problems**
- C) To avoid using data**
- D) To increase operational costs**

Answer: B) To simplify complex business problems

6. Models in BI primarily transform:

- A) Information into raw data**
- B) Data into meaningful insights**
- C) Decisions into confusion**
- D) Hardware into software**

Answer: B) Data into meaningful insights

7. In decision-making, models are most useful for:

- A) Guesswork**
- B) Emotional decisions**
- C) Structured and analytical evaluation**
- D) Ignoring risks**

Answer: C) Structured and analytical evaluation

8. Which of the following is NOT a benefit of using models in BI?

- A) Improved forecasting**
- B) Better resource allocation**
- C) Increased uncertainty**
- D) Enhanced strategic planning**

Answer: C) Increased uncertainty

9. A good BI model should be:

- A) Extremely complex and difficult to understand**
- B) Simple, relevant, and accurate**
- C) Based only on assumptions without data**
- D) Used only once**

Answer: B) Simple, relevant, and accurate

10. The role of models in decision-making includes:

- A) Eliminating all risks**
- B) Supporting scenario analysis and prediction**
- C) Avoiding future planning**

D) Ignoring business trends

Answer: B) Supporting scenario analysis and prediction